

Departamento	VLAN	Red / Prefijo	Máscara de Subred
Administración	10	10.1.10.0/24	255.255.255.0
Ventas	20	10.1.20.0/25	255.255.255.128
Logística	30	10.1.30.0/25	255.255.255.128
Invitados	40	10.1.40.0/26	255.255.255.192
Management	99	10.1.99.0/28	255.255.255.240

ID Túnel	Conexión (Sede A <--> Sede B)	IP Sede A	IP Sede B
T1	Of. Centrales <--> Suc. Norte	192.168.1.1	192.168.1.2
T2	Of. Centrales <--> Suc. Bajío	192.168.1.5	192.168.1.6
T3	Of. Centrales <--> Suc. Occidente	192.168.1.9	192.168.1.10
T4	Of. Centrales <--> Centro Dist.	192.168.1.13	192.168.1.14
T5	Suc. Norte <--> Suc. Bajío	192.168.1.17	192.168.1.18
T6	Suc. Norte <--> Suc. Occidente	192.168.1.21	192.168.1.22
T7	Suc. Norte <--> Centro Dist.	192.168.1.25	192.168.1.26
T8	Suc. Bajío <--> Suc. Occidente	192.168.1.29	192.168.1.30
T9	Suc. Bajío <--> Centro Dist.	192.168.1.33	192.168.1.34
T10	Suc. Occidente <--> Centro Dist.	192.168.1.37	192.168.1.38

Sede / Conexión	IP en tu Router (Sede)	IP en el Router ISP	Red del Enlace (/30)
Oficinas Centrales	200.10.20.1	200.10.20.2	200.10.20.0
Sucursal Norte	200.10.20.5	200.10.20.6	200.10.20.4
Sucursal Bajío	200.10.20.9	200.10.20.10	200.10.20.8
Sucursal Occidente	200.10.20.13	200.10.20.14	200.10.20.12
Centro de Distribución	200.10.20.17	200.10.20.18	200.10.20.16

ROuter isp

```
enable
config t

ip routing

interface vlan 3
 ip address 200.10.20.10 255.255.255.252
 ip ospf 1 area 0
 no shutdown
```

```

exit

interface vlan 4
 ip address 200.10.20.14 255.255.255.252
 ip ospf 1 area 0
 no shutdown
exit

interface vlan 5
 ip address 200.10.20.18 255.255.255.252
 ip ospf 1 area 0
 no shutdown
exit

interface GigabitEthernet0/0/1
 description enlace a Oficinas Centrales
 ip address 200.10.20.2 255.255.255.252
 ip ospf 1 area 0
 ip ospf network point-to-point
 no shutdown
exit

interface GigabitEthernet0/0/2
 description enlace a Sucursal Norte
 ip address 200.10.20.6 255.255.255.252
 ip ospf 1 area 0
 ip ospf network point-to-point
 no shutdown
exit

interface GigabitEthernet0/1/0
 description enlace a Sucursal Bajio
 switchport access vlan 3
 no shutdown
exit

interface GigabitEthernet0/1/1
 description enlace a Sucursal Occidente
 switchport access vlan 4
 no shutdown
exit

interface GigabitEthernet0/1/2
 description enlace a Sucursal Distribucion
 switchport access vlan 5
 no shutdown
exit

interface GigabitEthernet0/0/0
 ip address 8.8.8.1 255.255.255.0
exit

router ospf 1
 router-id 0.0.0.1
 network 200.10.20.2 0.0.0.0 area 0
 network 200.10.20.6 0.0.0.0 area 0
 network 200.10.20.10 0.0.0.0 area 0
 network 200.10.20.14 0.0.0.0 area 0
 network 200.10.20.18 0.0.0.0 area 0
 default-information originate

```

ofcinas centrales

router 0

```
enable
config terminal

ip access-list extended WEB
  deny tcp 10.1.20.0 0.0.0.127 host 8.8.8.10 eq www
  deny tcp 10.1.30.0 0.0.0.127 host 8.8.8.10 eq www
  deny tcp 10.1.40.0 0.0.0.63 host 8.8.8.10 eq www
  permit ip any any
exit

ip access-list extended Centrales-Norte
  permit ip 10.1.0.0 0.0.255.255 10.2.0.0 0.0.255.255
exit

ip access-list extended Centrales-Bajio
  permit ip 10.1.0.0 0.0.255.255 10.3.0.0 0.0.255.255
exit

ip access-list extended Centrales-Occidente
  permit ip 10.1.0.0 0.0.255.255 10.4.0.0 0.0.255.255
exit

ip access-list extended Centrales-Distribucion
  permit ip 10.1.0.0 0.0.255.255 10.5.0.0 0.0.255.255
exit

crypto isakmp policy 10
  encryption aes 256
  authentication pre-share
  group 5
exit

crypto isakmp key Cisco123 address 200.10.20.5
crypto isakmp key Cisco123 address 200.10.20.9
crypto isakmp key Cisco123 address 200.10.20.13
crypto isakmp key Cisco123 address 200.10.20.17

crypto ipsec transform-set TransData esp-aes 256 esp-sha-hmac

crypto map VPN-TransData 10 ipsec-isakmp
  set peer 200.10.20.5
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Centrales-Norte

crypto map VPN-TransData 20 ipsec-isakmp
  set peer 200.10.20.9
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
```

```

match address Centrales-Bajio

crypto map VPN-TransData 30 ipsec-isakmp
  set peer 200.10.20.13
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
match address Centrales-Occidente

crypto map VPN-TransData 40 ipsec-isakmp
  set peer 200.10.20.17
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
match address Centrales-Distribucion

interface GigabitEthernet 0/0/2
  ip address 200.10.20.1 255.255.255.252
  ip ospf 2 area 0
  ip ospf network point-to-point
  crypto map VPN-TransData
  ip nat outside
  no shutdown
exit

interface GigabitEthernet 0/0/0
  ip address 10.1.100.1 255.255.255.252
  ip ospf network point-to-point
  ip ospf 1 area 0
  ip access-group WEB in
  ip nat inside
  no shutdown
exit

interface GigabitEthernet 0/0/1
  ip address 10.1.100.5 255.255.255.252
  ip ospf network point-to-point
  ip ospf 1 area 0
  ip access-group WEB in
  ip nat inside
  no shutdown
exit

router ospf 1
  router-id 1.1.1.1
  default-information originate
exit

router ospf 2
  router-id 0.0.0.6
  network 200.10.20.0 0.0.0.3 area 0

ip access-list extended NAT-INTERNET
  deny ip 10.1.0.0 0.0.255.255 10.0.0.0 0.255.255.255
  permit ip 10.1.0.0 0.0.255.255 any

ip nat inside source list NAT-INTERNET interface GigabitEthernet0/0/2 overload

ip nat inside source static udp 10.1.10.10 123 200.10.20.1 123
ip nat inside source static udp 10.1.10.10 514 200.10.20.1 514

service timestamps log datetime msec
logging 10.1.10.10

```

sw capa 3

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
    name administracion
vlan 20
    name ventas
vlan 30
    name logistica
vlan 40
    name invitados
vlan 99
    name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
    channel-group 6 mode active
    no shutdown
exit

interface GigabitEthernet 1/0/3
    no switchport
    ip address 10.1.100.2 255.255.255.252
    ip ospf 1 area 0
    ip ospf network point-to-point
    no shutdown
exit

interface port-channel 6
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
    ip address 10.1.10.2 255.255.255.0
    standby 1 ip 10.1.10.1
    standby 1 priority 110
    standby 1 preempt
    ip ospf 1 area 0
    no shut

interface vlan 20
    ip address 10.1.20.2 255.255.255.128
    standby 2 ip 10.1.20.1
    standby 2 priority 110
    standby 2 preempt
    ip ospf 1 area 0
    no shut

interface vlan 30
    ip address 10.1.30.2 255.255.255.128
    standby 3 ip 10.1.30.1
    standby 3 priority 110
```

```

standby 3 preempt
ip ospf 1 area 0
no shut

interface vlan 40
ip address 10.1.40.2 255.255.255.192
standby 4 ip 10.1.40.1
standby 4 priority 110
standby 4 preempt
ip ospf 1 area 0
no shut

interface vlan 99
ip address 10.1.99.2 255.255.255.240
standby 5 ip 10.1.99.1
standby 5 priority 110
standby 5 preempt
ip ospf 1 area 0
no shut

router ospf 1
router-id 1.1.1.2
network 10.1.10.0 0.0.0.255 area 0
network 10.1.20.0 0.0.0.127 area 0
network 10.1.30.0 0.0.0.127 area 0
network 10.1.40.0 0.0.0.63 area 0
network 10.1.99.0 0.0.0.15 area 0
passive-interface vlan 10
passive-interface vlan 20
passive-interface vlan 30
passive-interface vlan 40
passive-interface vlan 99
exit

ip dhcp excluded-address 10.1.10.1 10.1.10.10
ip dhcp excluded-address 10.1.20.1 10.1.20.10
ip dhcp excluded-address 10.1.30.1 10.1.30.10
ip dhcp excluded-address 10.1.40.1 10.1.40.10
ip dhcp excluded-address 10.1.99.1 10.1.99.5
ip dhcp excluded-address 10.1.10.129 10.1.10.254
ip dhcp excluded-address 10.1.20.65 10.1.20.126
ip dhcp excluded-address 10.1.30.65 10.1.30.126
ip dhcp excluded-address 10.1.40.33 10.1.40.62
ip dhcp excluded-address 10.1.99.9 10.1.99.14

ip dhcp pool LAN-POOL-10
network 10.1.10.0 255.255.255.0
default-router 10.1.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.1.20.0 255.255.255.128
default-router 10.1.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.1.30.0 255.255.255.128
default-router 10.1.30.1
dns-server 8.8.8.8
domain-name cisco.com

```

```
exit
ip dhcp pool LAN-POOL-40
 network 10.1.40.0 255.255.255.192
 default-router 10.1.40.1
 dns-server 8.8.8.8
 domain-name cisco.com
exit
ip dhcp pool LAN-POOL-99
 network 10.1.99.0 255.255.255.240
 default-router 10.1.99.1
 dns-server 8.8.8.8
 domain-name cisco.com
exit
ntp server 10.1.10.10
service timestamps log datetime msec
logging 10.1.10.10
```

sw capa 3 (2) reserva

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
 name administracion
vlan 20
 name ventas
vlan 30
 name logistica
vlan 40
 name invitados
vlan 99
 name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
 channel-group 1 mode active
 no shutdown
exit

interface GigabitEthernet 1/0/3
 no switchport
 ip address 10.1.100.6 255.255.255.252
 ip ospf 1 area 0
 ip ospf network point-to-point
 no shutdown
exit

interface port-channel 1
 switchport mode trunk
 switchport nonegotiate
 switchport trunk allowed vlan 10,20,30,40,99
exit
```

```

interface vlan 10
 ip address 10.1.10.3 255.255.255.0
 standby 1 ip 10.1.10.1
 standby 1 priority 100
 standby 1 preempt
 ip ospf 1 area 0
 no shut

interface vlan 20
 ip address 10.1.20.3 255.255.255.128
 standby 2 ip 10.1.20.1
 standby 2 priority 100
 standby 2 preempt
 ip ospf 1 area 0
 no shut

interface vlan 30
 ip address 10.1.30.3 255.255.255.128
 standby 3 ip 10.1.30.1
 standby 3 priority 100
 standby 3 preempt
 ip ospf 1 area 0
 no shut

interface vlan 40
 ip address 10.1.40.3 255.255.255.192
 standby 4 ip 10.1.40.1
 standby 4 priority 100
 standby 4 preempt
 ip ospf 1 area 0
 no shut

interface vlan 99
 ip address 10.1.99.3 255.255.255.240
 standby 5 ip 10.1.99.1
 standby 5 priority 100
 standby 5 preempt
 ip ospf 1 area 0
 no shut

router ospf 1
 router-id 1.1.1.3
  network 10.1.10.0 0.0.0.255 area 0
  network 10.1.20.0 0.0.0.127 area 0
  network 10.1.30.0 0.0.0.127 area 0
  network 10.1.40.0 0.0.0.63 area 0
  network 10.1.99.0 0.0.0.15 area 0
  passive-interface vlan 10
  passive-interface vlan 20
  passive-interface vlan 30
  passive-interface vlan 40
  passive-interface vlan 99
  exit

ip dhcp excluded-address 10.1.10.1 10.1.10.128
ip dhcp excluded-address 10.1.20.1 10.1.20.64
ip dhcp excluded-address 10.1.30.1 10.1.30.64
ip dhcp excluded-address 10.1.40.1 10.1.40.32
ip dhcp excluded-address 10.1.99.1 10.1.99.8

ip dhcp pool LAN-POOL-10
 network 10.1.10.0 255.255.255.0
 default-router 10.1.10.1

```



```
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.1.20.0 255.255.255.128
default-router 10.1.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.1.30.0 255.255.255.128
default-router 10.1.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.1.40.0 255.255.255.192
default-router 10.1.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.1.99.0 255.255.255.240
default-router 10.1.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 10.1.10.10
service timestamps log datetime msec
logging 10.1.10.10
```

sw1

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
channel-group 6 mode active
```

```
exit

interface range fastEthernet 0/1-2
channel-group 1 mode active
exit

interface range fastEthernet 0/3-4
channel-group 2 mode active
exit

interface range fastEthernet 0/5-6
channel-group 3 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 2
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastethernet 0/7
description Acceso a Server
switchport mode access
switchport access vlan 10
switchport port-security
exit

interface fastEthernet 0/8
description Trunk a WLC
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fastEthernet 0/9-24
switchport mode access
switchport access vlan 1000
shutdown
exit
```

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range fastEthernet 0/1-2
  channel-group 1 mode active
  exit

interface range fastEthernet 0/7-8
  channel-group 4 mode active
  exit

interface range fastEthernet 0/11-12
  channel-group 6 mode active
  exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 4
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface fastEthernet 0/3
  description Trunk a AP
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  switchport trunk native vlan 99
  no shutdown
  exit

interface fastEthernet 0/4
  description Acceso a PC logistica
  switchport mode access
```

```
switchport access vlan 30
switchport port-security
no shutdown
exit
```

```
interface range fa0/5-6, fa0/9-10, fa0/13-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw3

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
channel-group 1 mode active
exit

interface range fastEthernet 0/5-6
channel-group 3 mode active
exit

interface range fastEthernet 0/9-10
channel-group 5 mode active
exit

interface range fastEthernet 0/11-12
channel-group 6 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
```

```

exit

interface port-channel 5
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/1
description Acceso a PC invitados
switchport mode access
switchport access vlan 40
switchport port-security
no shutdown
exit

interface fastEthernet 0/2
description Trunk a PC Management
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/3-4, fa0/7-8, fa0/13-24
switchport mode access
switchport access vlan 1000
shutdown
exit

```

sw4

```

enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

```

```
interface range FastEthernet 0/3-4
channel-group 2 mode active
exit

interface range FastEthernet 0/7-8
channel-group 4 mode active
exit

interface range FastEthernet 0/9-10
channel-group 5 mode active
exit

interface port-channel 2
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 5
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/1
description Acceso a PC administracion
switchport mode access
switchport access vlan 10
switchport port-security
no shutdown
exit

interface fastEthernet 0/2
description Acceso a PC ventas
switchport mode access
switchport access vlan 20
switchport port-security
no shutdown
exit

interface range fa0/5-6, fa0/11-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```

Sucursal norte

router

enable

config t

```
ip access-list extended ACL-CONTROL
  permit udp host 10.2.10.10 host 10.1.10.10 eq 69
  deny udp any host 10.1.10.10 eq 69
  deny tcp 10.2.20.0 0.0.0.127 host 8.8.8.10 eq www
  deny tcp 10.2.30.0 0.0.0.127 host 8.8.8.10 eq www
  deny tcp 10.2.40.0 0.0.0.63 host 8.8.8.10 eq www
  deny icmp 10.2.20.0 0.0.0.127 host 10.1.10.10
  deny icmp 10.2.30.0 0.0.0.127 host 10.1.10.10
  deny icmp 10.2.40.0 0.0.0.63 host 10.1.10.10
  permit ip any any
```

exit

```
ip access-list extended Norte-Centrales
  permit ip 10.2.0.0 0.0.255.255 10.1.0.0 0.0.255.255
```

exit

```
ip access-list extended Norte-Bajio
  permit ip 10.2.0.0 0.0.255.255 10.3.0.0 0.0.255.255
```

exit

```
ip access-list extended Norte-Occidente
  permit ip 10.2.0.0 0.0.255.255 10.4.0.0 0.0.255.255
```

exit

```
ip access-list extended Norte-Distribucion
  permit ip 10.2.0.0 0.0.255.255 10.5.0.0 0.0.255.255
```

exit

```
crypto isakmp policy 10
  encryption aes 256
  authentication pre-share
  group 5
  exit
```

```
crypto isakmp key Cisco123 address 200.10.20.1
crypto isakmp key Cisco123 address 200.10.20.9
crypto isakmp key Cisco123 address 200.10.20.13
crypto isakmp key Cisco123 address 200.10.20.17
```

```
crypto ipsec transform-set TransData esp-aes 256 esp-sha-hmac
```

```
crypto map VPN-TransData 10 ipsec-isakmp
  set peer 200.10.20.1
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Norte-Centrales
```

```
crypto map VPN-TransData 20 ipsec-isakmp
  set peer 200.10.20.9
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Norte-Bajio
```

```
crypto map VPN-TransData 30 ipsec-isakmp
  set peer 200.10.20.13
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Norte-Occidente
```

```

crypto map VPN-TransData 40 ipsec-isakmp
  set peer 200.10.20.17
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
match address Norte-Distribucion

interface GigabitEthernet 0/0/2
  ip address 200.10.20.5 255.255.255.252
  ip ospf 1 area 0
  ip ospf network point-to-point
  crypto map VPN-TransData
  ip nat outside
no shutdown
exit

interface GigabitEthernet 0/0/0
  ip address 10.2.100.1 255.255.255.252
  ip ospf network point-to-point
  ip ospf 2 area 0
  ip access-group ACL-CONTROL in
  ip nat inside
  no shutdown
exit

interface GigabitEthernet 0/0/1
  ip address 10.2.100.5 255.255.255.252
  ip ospf 2 area 0
  ip ospf network point-to-point
  ip access-group ACL-CONTROL in
  ip nat inside
  no shutdown
exit

router ospf 2
  router-id 1.1.2.1
  default-information originate

router ospf 1
  router-id 0.0.0.2
  network 200.10.20.4 0.0.0.3 area 0

ip access-list extended NAT-INTERNET
  deny ip 10.2.0.0 0.0.255.255 10.0.0.0 0.255.255.255
  permit ip 10.2.0.0 0.0.255.255 any

ip nat inside source list NAT-INTERNET interface GigabitEthernet0/0/2 overload

ntp server 200.10.20.1
service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo

snmp-server community NorteRO ro
snmp-server community NorteRW rw

```

sw capa 3

enable


```

configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
  channel-group 6 mode active
exit

interface GigabitEthernet 1/0/3
  no switchport
  ip address 10.2.100.2 255.255.255.252
  ip ospf 2 area 0
  ip ospf network point-to-point
  no shutdown
exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
  ip address 10.2.10.2 255.255.255.0
  standby 1 ip 10.2.10.1
  standby 1 priority 110
  standby 1 preempt
  ip ospf 2 area 0
  no shut

interface vlan 20
  ip address 10.2.20.2 255.255.255.128
  standby 2 ip 10.2.20.1
  standby 2 priority 110
  standby 2 preempt
  ip ospf 2 area 0
  no shut

interface vlan 30
  ip address 10.2.30.2 255.255.255.128
  standby 3 ip 10.2.30.1
  standby 3 priority 110
  standby 3 preempt
  ip ospf 2 area 0
  no shut

interface vlan 40
  ip address 10.2.40.2 255.255.255.192
  standby 4 ip 10.2.40.1

```

```

standby 4 priority 110
standby 4 preempt
ip ospf 2 area 0
no shut

interface vlan 99
ip address 10.2.99.2 255.255.255.240
standby 5 ip 10.2.99.1
standby 5 priority 110
standby 5 preempt
ip ospf 2 area 0
no shut

router ospf 2
router-id 1.1.2.2
network 10.2.10.0 0.0.0.255 area 0
network 10.2.20.0 0.0.0.127 area 0
network 10.2.30.0 0.0.0.127 area 0
network 10.2.40.0 0.0.0.63 area 0
network 10.2.99.0 0.0.0.15 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.2.10.1 10.2.10.10
ip dhcp excluded-address 10.2.20.1 10.2.20.10
ip dhcp excluded-address 10.2.30.1 10.2.30.10
ip dhcp excluded-address 10.2.40.1 10.2.40.10
ip dhcp excluded-address 10.2.99.1 10.2.99.5
ip dhcp excluded-address 10.2.10.129 10.2.10.254
ip dhcp excluded-address 10.2.20.65 10.2.20.126
ip dhcp excluded-address 10.2.30.65 10.2.30.126
ip dhcp excluded-address 10.2.40.33 10.2.40.62
ip dhcp excluded-address 10.2.99.9 10.2.99.14

ip dhcp pool LAN-POOL-10
network 10.2.10.0 255.255.255.0
default-router 10.2.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.2.20.0 255.255.255.128
default-router 10.2.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.2.30.0 255.255.255.128
default-router 10.2.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.2.40.0 255.255.255.192
default-router 10.2.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

```

```
ip dhcp pool LAN-POOL-99
 network 10.2.99.0 255.255.255.240
 default-router 10.2.99.1
 dns-server 8.8.8.8
 domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo
```

sw capa 3 (2) reserva

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
 name administracion
vlan 20
 name ventas
vlan 30
 name logistica
vlan 40
 name invitados
vlan 99
 name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
 channel-group 1 mode active
exit

interface GigabitEthernet 1/0/3
 no switchport
 ip address 10.2.100.6 255.255.255.252
 ip ospf 2 area 0
 ip ospf network point-to-point
 no shutdown
exit

interface port-channel 1
 switchport mode trunk
 switchport nonegotiate
 switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
 ip address 10.2.10.3 255.255.255.0
 standby 1 ip 10.2.10.1
 standby 1 priority 100
 standby 1 preempt
 ip ospf 2 area 0
```

```

no shut

interface vlan 20
ip address 10.2.20.3 255.255.255.128
standby 2 ip 10.2.20.1
standby 2 priority 100
standby 2 preempt
ip ospf 2 area 0
no shut

interface vlan 30
ip address 10.2.30.3 255.255.255.128
standby 3 ip 10.2.30.1
standby 3 priority 100
standby 3 preempt
ip ospf 2 area 0
no shut

interface vlan 40
ip address 10.2.40.3 255.255.255.192
standby 4 ip 10.2.40.1
standby 4 priority 100
standby 4 preempt
ip ospf 2 area 0
no shut

interface vlan 99
ip address 10.2.99.3 255.255.255.240
standby 5 ip 10.2.99.1
standby 5 priority 100
standby 5 preempt
ip ospf 2 area 0
no shut

router ospf 2
router-id 1.1.2.3
network 10.2.10.0 0.0.0.255 area 0
network 10.2.20.0 0.0.0.127 area 0
network 10.2.30.0 0.0.0.127 area 0
network 10.2.40.0 0.0.0.63 area 0
network 10.2.99.0 0.0.0.15 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.2.10.1 10.2.10.128
ip dhcp excluded-address 10.2.20.1 10.2.20.64
ip dhcp excluded-address 10.2.30.1 10.2.30.64
ip dhcp excluded-address 10.2.40.1 10.2.40.32
ip dhcp excluded-address 10.2.99.1 10.2.99.8

ip dhcp pool LAN-POOL-10
network 10.2.10.0 255.255.255.0
default-router 10.2.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.2.20.0 255.255.255.128
default-router 10.2.20.1

```

```
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.2.30.0 255.255.255.128
default-router 10.2.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.2.40.0 255.255.255.192
default-router 10.2.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.2.99.0 255.255.255.240
default-router 10.2.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo
```

sw5

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
channel-group 6 mode active
exit

interface range fastEthernet 0/1-2
channel-group 1 mode active
```

```
exit

interface range fastEthernet 0/3-4
channel-group 2 mode active
exit

interface range fastEthernet 0/5-6
channel-group 3 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 2
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastethernet 0/7
description Trunk a WLC
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/8-24
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw7

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
```

```

vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
  channel-group 1 mode active
  no shut
  exit

interface range fastEthernet 0/5-6
  channel-group 3 mode active
  exit

interface range fastEthernet 0/9-10
  channel-group 5 mode active
  exit

interface range fastEthernet 0/11-12
  channel-group 6 mode active
  exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 3
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 5
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface fastEthernet 0/1
  description Acceso a PC Administrador
  switchport mode access
  switchport access vlan 10
  switchport port-security
  no shutdown
  exit

interface range fa0/2-4, fa0/7-8, fa0/13-24
  switchport mode access
  switchport access vlan 1000
  shutdown
  exit

```

sw6

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
    name administracion
vlan 20
    name ventas
vlan 30
    name logistica
vlan 40
    name invitados
vlan 99
    name management
exit

vlan 1000
exit

interface range fastEthernet 0/1-2
    channel-group 1 mode active
exit

interface range fastEthernet 0/7-8
    channel-group 4 mode active
exit

interface range fastEthernet 0/11-12
    channel-group 6 mode active
exit

interface port-channel 1
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/3
    description Trunk a PC Management
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 10,20,30,40,99
    switchport trunk native vlan 99
    no shutdown
exit
```



```
interface range fa0/4-6, fa0/9-10, fa0/13-24, gi0/1-2
  switchport mode access
  switchport access vlan 1000
  shutdown
  exit
```

sw8

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
  exit

vlan 1000
  exit

interface range FastEthernet 0/3-4
  channel-group 2 mode active
  exit

interface range FastEthernet 0/7-8
  channel-group 4 mode active
  exit

interface range FastEthernet 0/9-10
  channel-group 5 mode active
  exit

interface port-channel 2
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 4
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 5
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit
```

```
interface fastEthernet 0/1
description Trunk a AP
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/2, fa0/5-6, fa0/11-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```

Sucursal Bajio

router

```
enable
config t

ip access-list extended ACL-CONTROL
permit udp host 10.3.10.10 host 10.1.10.10 eq 69
deny udp any host 10.1.10.10 eq 69
deny tcp 10.3.20.0 0.0.0.127 host 8.8.8.10 eq www
deny tcp 10.3.30.0 0.0.0.127 host 8.8.8.10 eq www
deny tcp 10.3.40.0 0.0.0.63 host 8.8.8.10 eq www
deny icmp 10.3.20.0 0.0.0.127 host 10.1.10.10
deny icmp 10.3.30.0 0.0.0.127 host 10.1.10.10
deny icmp 10.3.40.0 0.0.0.63 host 10.1.10.10
permit ip any any
exit

ip access-list extended Bajio-Centrales
permit ip 10.3.0.0 0.0.255.255 10.1.0.0 0.0.255.255
exit

ip access-list extended Bajio-Norte
permit ip 10.3.0.0 0.0.255.255 10.2.0.0 0.0.255.255
exit

ip access-list extended Bajio-Occidente
permit ip 10.3.0.0 0.0.255.255 10.4.0.0 0.0.255.255
exit

ip access-list extended Bajio-Distribucion
permit ip 10.3.0.0 0.0.255.255 10.5.0.0 0.0.255.255
exit

crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
exit
```

```

crypto isakmp key Cisco123 address 200.10.20.1
crypto isakmp key Cisco123 address 200.10.20.5
crypto isakmp key Cisco123 address 200.10.20.13
crypto isakmp key Cisco123 address 200.10.20.17

crypto ipsec transform-set TransData esp-aes 256 esp-sha-hmac

crypto map VPN-TransData 10 ipsec-isakmp
  set peer 200.10.20.1
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Bajio-Centrales

crypto map VPN-TransData 20 ipsec-isakmp
  set peer 200.10.20.5
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Bajio-Norte

crypto map VPN-TransData 30 ipsec-isakmp
  set peer 200.10.20.13
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Bajio-Occidente

crypto map VPN-TransData 40 ipsec-isakmp
  set peer 200.10.20.17
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Bajio-Distribucion

interface gigabitEthernet 0/0/2
  ip address 200.10.20.9 255.255.255.252
  ip ospf 1 area 0
  ip ospf network point-to-point
  crypto map VPN-TransData
  ip nat outside
  no shutdown
  exit

interface GigabitEthernet 0/0/0
  ip address 10.3.100.1 255.255.255.252
  ip ospf network point-to-point
  ip ospf 3 area 0
  ip access-group ACL-CONTROL in
  ip nat inside
  no shutdown
  exit

interface GigabitEthernet 0/0/1
  ip address 10.3.100.5 255.255.255.252
  ip ospf 3 area 0
  ip ospf network point-to-point
  ip access-group ACL-CONTROL in
  ip nat inside
  no shutdown
  exit

router ospf 3
  router-id 1.1.3.1

```

```
default-information originate

router ospf 1
  router-id 0.0.0.3
  network 200.10.20.8 0.0.0.3 area 0

ip access-list extended NAT-INTERNET
  deny ip 10.3.0.0 0.0.255.255 10.0.0.0 0.255.255.255
  permit ip 10.3.0.0 0.0.255.255 any

ip nat inside source list NAT-INTERNET interface GigabitEthernet0/0/2 overload

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo

snmp-server community BajioRO ro
snmp-server community BajioRW rw
```

sw capa 3

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
  channel-group 6 mode active
exit

interface GigabitEthernet 1/0/3
  no switchport
  ip address 10.3.100.2 255.255.255.252
  ip ospf 3 area 0
  ip ospf network point-to-point
  no shutdown
exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit
```

```

interface vlan 10
 ip address 10.3.10.2 255.255.255.0
 standby 1 ip 10.3.10.1
 standby 1 priority 110
 standby 1 preempt
 ip ospf 3 area 0
 no shut

interface vlan 20
 ip address 10.3.20.2 255.255.255.128
 standby 2 ip 10.3.20.1
 standby 2 priority 110
 standby 2 preempt
 ip ospf 3 area 0
 no shut

interface vlan 30
 ip address 10.3.30.2 255.255.255.128
 standby 3 ip 10.3.30.1
 standby 3 priority 110
 standby 3 preempt
 ip ospf 3 area 0
 no shut

interface vlan 40
 ip address 10.3.40.2 255.255.255.192
 standby 4 ip 10.3.40.1
 standby 4 priority 110
 standby 4 preempt
 ip ospf 3 area 0
 no shut

interface vlan 99
 ip address 10.3.99.2 255.255.255.240
 standby 5 ip 10.3.99.1
 standby 5 priority 110
 standby 5 preempt
 ip ospf 3 area 0
 no shut

router ospf 3
 router-id 1.1.3.2
 network 10.3.10.0 0.0.0.255 area 0
 network 10.3.20.0 0.0.0.127 area 0
 network 10.3.30.0 0.0.0.127 area 0
 network 10.3.40.0 0.0.0.63 area 0
 network 10.3.99.0 0.0.0.15 area 0
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
 passive-interface Vlan99
 exit

ip dhcp excluded-address 10.3.10.1 10.3.10.10
ip dhcp excluded-address 10.3.20.1 10.3.20.10
ip dhcp excluded-address 10.3.30.1 10.3.30.10
ip dhcp excluded-address 10.3.40.1 10.3.40.10
ip dhcp excluded-address 10.3.99.1 10.3.99.5
ip dhcp excluded-address 10.3.10.129 10.3.10.254
ip dhcp excluded-address 10.3.20.65 10.3.20.126
ip dhcp excluded-address 10.3.30.65 10.3.30.126

```

```
ip dhcp excluded-address 10.3.40.33 10.3.40.62
ip dhcp excluded-address 10.3.99.9 10.3.99.14

ip dhcp pool LAN-POOL-10
network 10.3.10.0 255.255.255.0
default-router 10.3.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.3.20.0 255.255.255.128
default-router 10.3.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.3.30.0 255.255.255.128
default-router 10.3.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.3.40.0 255.255.255.192
default-router 10.3.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.3.99.0 255.255.255.240
default-router 10.3.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo
```

sw capa 3 (2) reserva

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
```

```
vlan 99
  name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
  channel-group 1 mode active
exit

interface GigabitEthernet 1/0/3
  no switchport
  ip address 10.3.100.6 255.255.255.252
  ip ospf 3 area 0
  ip ospf network point-to-point
  no shutdown
exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
  ip address 10.3.10.3 255.255.255.0
  standby 1 ip 10.3.10.1
  standby 1 priority 100
  standby 1 preempt
  ip ospf 3 area 0
  no shut

interface vlan 20
  ip address 10.3.20.3 255.255.255.128
  standby 2 ip 10.3.20.1
  standby 2 priority 100
  standby 2 preempt
  ip ospf 3 area 0
  no shut

interface vlan 30
  ip address 10.3.30.3 255.255.255.128
  standby 3 ip 10.3.30.1
  standby 3 priority 100
  standby 3 preempt
  ip ospf 3 area 0
  no shut

interface vlan 40
  ip address 10.3.40.3 255.255.255.192
  standby 4 ip 10.3.40.1
  standby 4 priority 100
  standby 4 preempt
  ip ospf 3 area 0
  no shut

interface vlan 99
  ip address 10.3.99.3 255.255.255.240
  standby 5 ip 10.3.99.1
  standby 5 priority 100
  standby 5 preempt
  ip ospf 3 area 0
  no shut

router ospf 3
```

```
router-id 1.1.3.3
network 10.3.10.0 0.0.0.255 area 0
network 10.3.20.0 0.0.0.127 area 0
network 10.3.30.0 0.0.0.127 area 0
network 10.3.40.0 0.0.0.63 area 0
network 10.3.99.0 0.0.0.15 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.3.10.1 10.3.10.128
ip dhcp excluded-address 10.3.20.1 10.3.20.64
ip dhcp excluded-address 10.3.30.1 10.3.30.64
ip dhcp excluded-address 10.3.40.1 10.3.40.32
ip dhcp excluded-address 10.3.99.1 10.3.99.8

ip dhcp pool LAN-POOL-10
network 10.3.10.0 255.255.255.0
default-router 10.3.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.3.20.0 255.255.255.128
default-router 10.3.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.3.30.0 255.255.255.128
default-router 10.3.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.3.40.0 255.255.255.192
default-router 10.3.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.3.99.0 255.255.255.240
default-router 10.3.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo
```



```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
  channel-group 6 mode active
exit

interface range fastEthernet 0/1-2
  channel-group 1 mode active
exit

interface range fastEthernet 0/3-4
  channel-group 2 mode active
exit

interface range fastEthernet 0/5-6
  channel-group 3 mode active
exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 2
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastethernet 0/7
  description Trunk a WLC
```

```
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/8-24
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw10

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range fastEthernet 0/1-2
channel-group 1 mode active
exit

interface range fastEthernet 0/7-8
channel-group 4 mode active
exit

interface range fastEthernet 0/11-12
channel-group 6 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
```

```
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/3
description Trunk a PC Management
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/4-6, fa0/9-10, fa0/13-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw11

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
 name administracion
vlan 20
 name ventas
vlan 30
 name logistica
vlan 40
 name invitados
vlan 99
 name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
channel-group 1 mode active
no shut
exit

interface range fastEthernet 0/5-6
channel-group 3 mode active
exit

interface range fastEthernet 0/9-10
channel-group 5 mode active
exit

interface range fastEthernet 0/11-12
channel-group 6 mode active
exit

interface port-channel 1
```

```
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 5
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/1
description Acceso a PC Administrador
switchport mode access
switchport access vlan 10
switchport port-security
no shutdown
exit

interface range fa0/2-4, fa0/7-8, fa0/13-24
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw12

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
 name administracion
vlan 20
 name ventas
vlan 30
 name logistica
vlan 40
 name invitados
vlan 99
 name management
exit

vlan 1000
exit
```

```

interface range FastEthernet 0/3-4
channel-group 2 mode active
exit

interface range FastEthernet 0/7-8
channel-group 4 mode active
exit

interface range FastEthernet 0/9-10
channel-group 5 mode active
exit

interface port-channel 2
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 5
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/1
description Trunk a AP
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/2, fa0/5-6, fa0/11-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit

```

Sucursal Occidente

router

```

enable
config terminal

ip access-list extended ACL-CONTROL
permit udp host 10.4.10.10 host 10.1.10.10 eq 69
deny udp any host 10.1.10.10 eq 69
deny tcp 10.4.20.0 0.0.0.127 host 8.8.8.10 eq www

```

```

deny tcp 10.4.30.0 0.0.0.127 host 8.8.8.10 eq www
deny tcp 10.4.40.0 0.0.0.63 host 8.8.8.10 eq www
deny icmp 10.4.20.0 0.0.0.127 host 10.1.10.10
deny icmp 10.4.30.0 0.0.0.127 host 10.1.10.10
deny icmp 10.4.40.0 0.0.0.63 host 10.1.10.10
permit ip any any
exit

ip access-list extended Occidente-Centrales
permit ip 10.4.0.0 0.0.255.255 10.1.0.0 0.0.255.255
exit

ip access-list extended Occidente-Norte
permit ip 10.4.0.0 0.0.255.255 10.2.0.0 0.0.255.255
exit

ip access-list extended Occidente-Bajio
permit ip 10.4.0.0 0.0.255.255 10.3.0.0 0.0.255.255
exit

ip access-list extended Occidente-Distribucion
permit ip 10.4.0.0 0.0.255.255 10.5.0.0 0.0.255.255
exit

crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
exit

crypto isakmp key Cisco123 address 200.10.20.1
crypto isakmp key Cisco123 address 200.10.20.5
crypto isakmp key Cisco123 address 200.10.20.9
crypto isakmp key Cisco123 address 200.10.20.17

crypto ipsec transform-set TransData esp-aes 256 esp-sha-hmac

crypto map VPN-TransData 10 ipsec-isakmp
set peer 200.10.20.1
set pfs group5
set security-association lifetime seconds 86400
set transform-set TransData
match address Occidente-Centrales

crypto map VPN-TransData 20 ipsec-isakmp
set peer 200.10.20.5
set pfs group5
set security-association lifetime seconds 86400
set transform-set TransData
match address Occidente-Norte

crypto map VPN-TransData 30 ipsec-isakmp
set peer 200.10.20.9
set pfs group5
set security-association lifetime seconds 86400
set transform-set TransData
match address Occidente-Bajio

crypto map VPN-TransData 40 ipsec-isakmp
set peer 200.10.20.17
set pfs group5
set security-association lifetime seconds 86400
set transform-set TransData
match address Occidente-Distribucion

```

```

interface gigabitEthernet 0/0/2
 ip address 200.10.20.13 255.255.255.252
 ip ospf 1 area 0
 ip ospf network point-to-point
 crypto map VPN-TransData
 ip nat outside
 no shutdown
exit

interface GigabitEthernet 0/0/0
 ip address 10.4.100.1 255.255.255.252
 ip ospf network point-to-point
 ip ospf 4 area 0
 ip access-group ACL-CONTROL in
 ip nat inside
 no shutdown
exit

interface GigabitEthernet 0/0/1
 ip address 10.4.100.5 255.255.255.252
 ip ospf 4 area 0
 ip ospf network point-to-point
 ip access-group ACL-CONTROL in
 ip nat inside
 no shutdown
exit

router ospf 4
 router-id 1.1.4.1
 default-information originate

router ospf 1
 router-id 0.0.0.4
 network 200.10.20.12 0.0.0.3 area 0

ip access-list extended NAT-INTERNET
 deny ip 10.4.0.0 0.0.255.255 10.0.0.0 0.255.255.255
 permit ip 10.4.0.0 0.0.255.255 any

ip nat inside source list NAT-INTERNET interface GigabitEthernet0/0/2 overload

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo

snmp-server community OccidenteRO ro
snmp-server community OccidenteRW rw

```

sw capa 3

```

enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
 name administracion

```

```
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
  channel-group 6 mode active
exit

interface GigabitEthernet 1/0/3
  no switchport
  ip address 10.4.100.2 255.255.255.252
  ip ospf 4 area 0
  ip ospf network point-to-point
  no shutdown
exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
  ip address 10.4.10.2 255.255.255.0
  standby 1 ip 10.4.10.1
  standby 1 priority 110
  standby 1 preempt
  ip ospf 4 area 0
  no shut

interface vlan 20
  ip address 10.4.20.2 255.255.255.128
  standby 2 ip 10.4.20.1
  standby 2 priority 110
  standby 2 preempt
  ip ospf 4 area 0
  no shut

interface vlan 30
  ip address 10.4.30.2 255.255.255.128
  standby 3 ip 10.4.30.1
  standby 3 priority 110
  standby 3 preempt
  ip ospf 4 area 0
  no shut

interface vlan 40
  ip address 10.4.40.2 255.255.255.192
  standby 4 ip 10.4.40.1
  standby 4 priority 110
  standby 4 preempt
  ip ospf 4 area 0
  no shut

interface vlan 99
  ip address 10.4.99.2 255.255.255.240
```



```
standby 5 ip 10.4.99.1
standby 5 priority 110
standby 5 preempt
ip ospf 4 area 0
no shut

router ospf 4
router-id 1.1.4.2
network 10.4.10.0 0.0.0.255 area 0
network 10.4.20.0 0.0.0.127 area 0
network 10.4.30.0 0.0.0.127 area 0
network 10.4.40.0 0.0.0.63 area 0
network 10.4.99.0 0.0.0.15 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.4.10.1 10.4.10.10
ip dhcp excluded-address 10.4.20.1 10.4.20.10
ip dhcp excluded-address 10.4.30.1 10.4.30.10
ip dhcp excluded-address 10.4.40.1 10.4.40.10
ip dhcp excluded-address 10.4.99.1 10.4.99.5
ip dhcp excluded-address 10.4.10.129 10.4.10.254
ip dhcp excluded-address 10.4.20.65 10.4.20.126
ip dhcp excluded-address 10.4.30.65 10.4.30.126
ip dhcp excluded-address 10.4.40.33 10.4.40.62
ip dhcp excluded-address 10.4.99.9 10.4.99.14

ip dhcp pool LAN-POOL-10
network 10.4.10.0 255.255.255.0
default-router 10.4.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.4.20.0 255.255.255.128
default-router 10.4.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.4.30.0 255.255.255.128
default-router 10.4.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.4.40.0 255.255.255.192
default-router 10.4.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.4.99.0 255.255.255.240
default-router 10.4.99.1
dns-server 8.8.8.8
domain-name cisco.com
```

```
exit
```

```
ntp server 200.10.20.1
```

```
service timestamps log datetime msec  
logging host 200.10.20.1  
logging trap debugging  
logging userinfo
```

sw capa 3 (2) reserva

```
enable
```

```
configure terminal
```

```
spanning-tree portfast default  
spanning-tree portfast bpduguard default
```

```
vlan 10  
    name administracion  
vlan 20  
    name ventas  
vlan 30  
    name logistica  
vlan 40  
    name invitados  
vlan 99  
    name management
```

```
exit
```

```
ip routing
```

```
interface range GigabitEthernet 1/0/1-2  
    channel-group 1 mode active
```

```
exit
```

```
interface GigabitEthernet 1/0/3  
    no switchport  
    ip address 10.4.100.6 255.255.255.252  
    ip ospf 4 area 0  
    ip ospf network point-to-point  
    no shutdown
```

```
exit
```

```
interface port-channel 1  
    switchport mode trunk  
    switchport nonegotiate  
    switchport trunk allowed vlan 10,20,30,40,99
```

```
exit
```

```
interface vlan 10  
    ip address 10.4.10.3 255.255.255.0  
    standby 1 ip 10.4.10.1  
    standby 1 priority 100  
    standby 1 preempt  
    ip ospf 4 area 0  
    no shut
```

```
interface vlan 20  
    ip address 10.4.20.3 255.255.255.128  
    standby 2 ip 10.4.20.1  
    standby 2 priority 100
```

```

standby 2 preempt
ip ospf 4 area 0
no shut

interface vlan 30
ip address 10.4.30.3 255.255.255.128
standby 3 ip 10.4.30.1
standby 3 priority 100
standby 3 preempt
ip ospf 4 area 0
no shut

interface vlan 40
ip address 10.4.40.3 255.255.255.192
standby 4 ip 10.4.40.1
standby 4 priority 100
standby 4 preempt
ip ospf 4 area 0
no shut

interface vlan 99
ip address 10.4.99.3 255.255.255.240
standby 5 ip 10.4.99.1
standby 5 priority 100
standby 5 preempt
ip ospf 4 area 0
no shut

router ospf 4
router-id 1.1.4.3
network 10.4.10.0 0.0.0.255 area 0
network 10.4.20.0 0.0.0.127 area 0
network 10.4.30.0 0.0.0.127 area 0
network 10.4.40.0 0.0.0.63 area 0
network 10.4.99.0 0.0.0.15 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.4.10.1 10.4.10.128
ip dhcp excluded-address 10.4.20.1 10.4.20.64
ip dhcp excluded-address 10.4.30.1 10.4.30.64
ip dhcp excluded-address 10.4.40.1 10.4.40.32
ip dhcp excluded-address 10.4.99.1 10.4.99.8

ip dhcp pool LAN-POOL-10
network 10.4.10.0 255.255.255.0
default-router 10.4.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.4.20.0 255.255.255.128
default-router 10.4.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.4.30.0 255.255.255.128
default-router 10.4.30.1

```

```
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.4.40.0 255.255.255.192
default-router 10.4.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.4.99.0 255.255.255.240
default-router 10.4.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo
```

sw13

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
channel-group 6 mode active
exit

interface range fastEthernet 0/1-2
channel-group 1 mode active
exit

interface range fastEthernet 0/3-4
channel-group 2 mode active
exit

interface range fastEthernet 0/5-6
```

```
channel-group 3 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 2
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastethernet 0/7
description Trunk a WLC
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/8-24
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw14

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit
```

```
vlan 1000
exit

interface range fastEthernet 0/1-2
channel-group 1 mode active
exit

interface range fastEthernet 0/7-8
channel-group 4 mode active
exit

interface range fastEthernet 0/11-12
channel-group 6 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/3
description Trunk a PC Management
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/4-6, fa0/9-10, fa0/13-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw15

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
```

```

vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
  channel-group 1 mode active
  no shut
  exit

interface range fastEthernet 0/5-6
  channel-group 3 mode active
  exit

interface range fastEthernet 0/9-10
  channel-group 5 mode active
  exit

interface range fastEthernet 0/11-12
  channel-group 6 mode active
  exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 3
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 5
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface fastEthernet 0/1
  description Acceso a PC Administrador
  switchport mode access
  switchport access vlan 10
  switchport port-security
  no shutdown
  exit

interface range fa0/2-4, fa0/7-8, fa0/13-24
  switchport mode access
  switchport access vlan 1000
  shutdown
  exit

```

sw16

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range FastEthernet 0/3-4
  channel-group 2 mode active
  exit

interface range FastEthernet 0/7-8
  channel-group 4 mode active
  exit

interface range FastEthernet 0/9-10
  channel-group 5 mode active
  exit

interface port-channel 2
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 4
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface port-channel 5
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  exit

interface fastEthernet 0/1
  description Trunk a AP
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  switchport trunk native vlan 99
  no shutdown
```



```
exit
```

```
interface range fa0/2, fa0/5-6, fa0/11-24, gi0/1-2
 switchport mode access
 switchport access vlan 1000
 shutdown
exit
```

Centro Distribucion

router

```
enable
```

```
config terminal
```

```
ip access-list extended ACL-CONTROL
 permit udp host 10.5.10.10 host 10.1.10.10 eq 69
 deny udp any host 10.1.10.10 eq 69
 deny tcp 10.5.20.0 0.0.0.127 host 8.8.8.10 eq www
 deny tcp 10.5.30.0 0.0.0.127 host 8.8.8.10 eq www
 deny tcp 10.5.40.0 0.0.0.63 host 8.8.8.10 eq www
 deny icmp 10.5.20.0 0.0.0.127 host 10.1.10.10
 deny icmp 10.5.30.0 0.0.0.127 host 10.1.10.10
 deny icmp 10.5.40.0 0.0.0.63 host 10.1.10.10
 permit ip any any
```

```
exit
```

```
ip access-list extended Distribucion-Centrales
 permit ip 10.5.0.0 0.0.255.255 10.1.0.0 0.0.255.255
```

```
exit
```

```
ip access-list extended Distribucion-Norte
 permit ip 10.5.0.0 0.0.255.255 10.2.0.0 0.0.255.255
```

```
exit
```

```
ip access-list extended Distribucion-Bajio
 permit ip 10.5.0.0 0.0.255.255 10.3.0.0 0.0.255.255
```

```
exit
```

```
ip access-list extended Distribucion-Occidente
 permit ip 10.5.0.0 0.0.255.255 10.4.0.0 0.0.255.255
```

```
exit
```

```
crypto isakmp policy 10
 encryption aes 256
 authentication pre-share
 group 5
exit
```

```
crypto isakmp key Cisco123 address 200.10.20.1
crypto isakmp key Cisco123 address 200.10.20.5
crypto isakmp key Cisco123 address 200.10.20.9
crypto isakmp key Cisco123 address 200.10.20.13
```

```
crypto ipsec transform-set TransData esp-aes 256 esp-sha-hmac
```

```

crypto map VPN-TransData 10 ipsec-isakmp
  set peer 200.10.20.1
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Distribucion-Centrales

crypto map VPN-TransData 20 ipsec-isakmp
  set peer 200.10.20.5
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Distribucion-Norte

crypto map VPN-TransData 30 ipsec-isakmp
  set peer 200.10.20.9
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Distribucion-Bajio

crypto map VPN-TransData 40 ipsec-isakmp
  set peer 200.10.20.13
  set pfs group5
  set security-association lifetime seconds 86400
  set transform-set TransData
  match address Distribucion-Occidente

interface gigabitEthernet 0/0/2
ip address 200.10.20.17 255.255.255.252
ip ospf 1 area 0
ip ospf network point-to-point
crypto map VPN-TransData
ip nat outside
no shutdown
exit

interface GigabitEthernet 0/0/0
  ip address 10.5.100.1 255.255.255.252
  ip ospf network point-to-point
  ip ospf 5 area 0
  ip access-group ACL-CONTROL in
  ip nat inside
  no shutdown
exit

interface GigabitEthernet 0/0/1
  ip address 10.5.100.5 255.255.255.252
  ip ospf 5 area 0
  ip ospf network point-to-point
  ip access-group ACL-CONTROL in
  ip nat inside
  no shutdown
exit

router ospf 5
  router-id 1.1.5.1
  default-information originate

router ospf 1
  router-id 0.0.0.5
  network 200.10.20.16 0.0.0.3 area 0

ip access-list extended NAT-INTERNET

```

```
deny ip 10.5.0.0 0.0.255.255 10.0.0.0 0.255.255.255
permit ip 10.5.0.0 0.0.255.255 any

ip nat inside source list NAT-INTERNET interface GigabitEthernet0/0/2 overload

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo

snmp-server community DistribucionRO ro
snmp-server community DistribucionRW rw
```

sw capa 3

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
 name administracion
vlan 20
 name ventas
vlan 30
 name logistica
vlan 40
 name invitados
vlan 99
 name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
 channel-group 6 mode active
exit

interface GigabitEthernet 1/0/3
 no switchport
 ip address 10.5.100.2 255.255.255.252
 ip ospf 5 area 0
 ip ospf network point-to-point
 no shutdown
exit

interface port-channel 6
 switchport mode trunk
 switchport nonegotiate
 switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
 ip address 10.5.10.2 255.255.255.0
 standby 1 ip 10.5.10.1
 standby 1 priority 110
 standby 1 preempt
```

```

ip ospf 5 area 0
no shut

interface vlan 20
ip address 10.5.20.2 255.255.255.128
standby 2 ip 10.5.20.1
standby 2 priority 110
standby 2 preempt
ip ospf 5 area 0
no shut

interface vlan 30
ip address 10.5.30.2 255.255.255.128
standby 3 ip 10.5.30.1
standby 3 priority 110
standby 3 preempt
ip ospf 5 area 0
no shut

interface vlan 40
ip address 10.5.40.2 255.255.255.192
standby 4 ip 10.5.40.1
standby 4 priority 110
standby 4 preempt
ip ospf 5 area 0
no shut

interface vlan 99
ip address 10.5.99.2 255.255.255.240
standby 5 ip 10.5.99.1
standby 5 priority 110
standby 5 preempt
ip ospf 5 area 0
no shut

router ospf 5
router-id 1.1.5.2
network 10.5.10.0 0.0.0.255 area 0
network 10.5.20.0 0.0.0.127 area 0
network 10.5.30.0 0.0.0.127 area 0
network 10.5.40.0 0.0.0.63 area 0
network 10.5.99.0 0.0.0.15 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.5.10.1 10.5.10.10
ip dhcp excluded-address 10.5.20.1 10.5.20.10
ip dhcp excluded-address 10.5.30.1 10.5.30.10
ip dhcp excluded-address 10.5.40.1 10.5.40.10
ip dhcp excluded-address 10.5.99.1 10.5.99.5
ip dhcp excluded-address 10.5.10.129 10.5.10.254
ip dhcp excluded-address 10.5.20.65 10.5.20.126
ip dhcp excluded-address 10.5.30.65 10.5.30.126
ip dhcp excluded-address 10.5.40.33 10.5.40.62
ip dhcp excluded-address 10.5.99.9 10.5.99.14

ip dhcp pool LAN-POOL-10
network 10.5.10.0 255.255.255.0
default-router 10.5.10.1
dns-server 8.8.8.8

```

```
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.5.20.0 255.255.255.128
default-router 10.5.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.5.30.0 255.255.255.128
default-router 10.5.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.5.40.0 255.255.255.192
default-router 10.5.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.5.99.0 255.255.255.240
default-router 10.5.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo
```

sw capa 3 (2) reserva

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

ip routing

interface range GigabitEthernet 1/0/1-2
```

```

channel-group 1 mode active
exit

interface GigabitEthernet 1/0/3
no switchport
ip address 10.5.100.6 255.255.255.252
ip ospf 5 area 0
ip ospf network point-to-point
no shutdown
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface vlan 10
ip address 10.5.10.3 255.255.255.0
standby 1 ip 10.5.10.1
standby 1 priority 100
standby 1 preempt
ip ospf 5 area 0
no shut

interface vlan 20
ip address 10.5.20.3 255.255.255.128
standby 2 ip 10.5.20.1
standby 2 priority 100
standby 2 preempt
ip ospf 5 area 0
no shut

interface vlan 30
ip address 10.5.30.3 255.255.255.128
standby 3 ip 10.5.30.1
standby 3 priority 100
standby 3 preempt
ip ospf 5 area 0
no shut

interface vlan 40
ip address 10.5.40.3 255.255.255.192
standby 4 ip 10.5.40.1
standby 4 priority 100
standby 4 preempt
ip ospf 5 area 0
no shut

interface vlan 99
ip address 10.5.99.3 255.255.255.240
standby 5 ip 10.5.99.1
standby 5 priority 100
standby 5 preempt
ip ospf 5 area 0
no shut

router ospf 5
router-id 1.1.5.3
network 10.5.10.0 0.0.0.255 area 0
network 10.5.20.0 0.0.0.127 area 0
network 10.5.30.0 0.0.0.127 area 0
network 10.5.40.0 0.0.0.63 area 0
network 10.5.99.0 0.0.0.15 area 0
passive-interface Vlan10

```

```
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan99
exit

ip dhcp excluded-address 10.5.10.1 10.5.10.128
ip dhcp excluded-address 10.5.20.1 10.5.20.64
ip dhcp excluded-address 10.5.30.1 10.5.30.64
ip dhcp excluded-address 10.5.40.1 10.5.40.32
ip dhcp excluded-address 10.5.99.1 10.5.99.8

ip dhcp pool LAN-POOL-10
network 10.5.10.0 255.255.255.0
default-router 10.5.10.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-20
network 10.5.20.0 255.255.255.128
default-router 10.5.20.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-30
network 10.5.30.0 255.255.255.128
default-router 10.5.30.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-40
network 10.5.40.0 255.255.255.192
default-router 10.5.40.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ip dhcp pool LAN-POOL-99
network 10.5.99.0 255.255.255.240
default-router 10.5.99.1
dns-server 8.8.8.8
domain-name cisco.com
exit

ntp server 200.10.20.1

service timestamps log datetime msec
logging host 200.10.20.1
logging trap debugging
logging userinfo

snmp-server community CentralesRO ro
snmp-server community CentralesRW rw
```

sw17

```
enable
configure terminal
```

```

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
  channel-group 6 mode active
exit

interface range fastEthernet 0/1-2
  channel-group 1 mode active
exit

interface range fastEthernet 0/3-4
  channel-group 2 mode active
exit

interface range fastEthernet 0/5-6
  channel-group 3 mode active
exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 2
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 3
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastethernet 0/7
  description Trunk a WLC
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
  switchport trunk native vlan 99

```



```
no shutdown
exit

interface range fa0/8-24
  switchport mode access
  switchport access vlan 1000
  shutdown
exit
```

sw18

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
  name administracion
vlan 20
  name ventas
vlan 30
  name logistica
vlan 40
  name invitados
vlan 99
  name management
exit

vlan 1000
exit

interface range fastEthernet 0/1-2
  channel-group 1 mode active
exit

interface range fastEthernet 0/7-8
  channel-group 4 mode active
exit

interface range fastEthernet 0/11-12
  channel-group 6 mode active
exit

interface port-channel 1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 10,20,30,40,99
exit
```

```
interface fastEthernet 0/3
description Trunk a PC Management
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/4-6, fa0/9-10, fa0/13-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```

sw19

```
enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range GigabitEthernet 0/1-2
channel-group 1 mode active
no shut
exit

interface range fastEthernet 0/5-6
channel-group 3 mode active
exit

interface range fastEthernet 0/9-10
channel-group 5 mode active
exit

interface range fastEthernet 0/11-12
channel-group 6 mode active
exit

interface port-channel 1
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit
```

```

interface port-channel 3
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 5
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 6
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/1
description Acceso a PC Administrador
switchport mode access
switchport access vlan 10
switchport port-security
no shutdown
exit

interface range fa0/2-4, fa0/7-8, fa0/13-24
switchport mode access
switchport access vlan 1000
shutdown
exit

```

sw20

```

enable
configure terminal

spanning-tree portfast default
spanning-tree portfast bpduguard default

vlan 10
name administracion
vlan 20
name ventas
vlan 30
name logistica
vlan 40
name invitados
vlan 99
name management
exit

vlan 1000
exit

interface range FastEthernet 0/3-4
channel-group 2 mode active
exit

```

```
interface range FastEthernet 0/7-8
channel-group 4 mode active
exit

interface range FastEthernet 0/9-10
channel-group 5 mode active
exit

interface port-channel 2
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 4
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface port-channel 5
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
exit

interface fastEthernet 0/1
description Trunk a AP
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 10,20,30,40,99
switchport trunk native vlan 99
no shutdown
exit

interface range fa0/2, fa0/5-6, fa0/11-24, gi0/1-2
switchport mode access
switchport access vlan 1000
shutdown
exit
```